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CS3002 Information Security



Introduction to Information Security

Course Outline



- Please refer to course outline for details of CLOs, evaluations and topic schedule



What Is Security?

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- “A state of being secure and free from danger or harm; the actions taken to make someone or something secure.”



What Is Information Security?

What Is Security?



- “A state of being secure and free from danger or harm; the actions taken to make someone or something secure.”

Information Security

- Protection of information and the systems that use, store, and transmit it from unauthorized access or usage, disruption, modification, or destruction.

Components of an Information System



Information system (IS) is the entire set of people, procedures, and technology that enable business to use information.

- Software
 - most difficult to secure
 - easy target
- Hardware
 - require physical security policies

Components of an Information System



- Data
 - Often most valuable asset
 - Main target of intentional attacks
- People
 - Weakest link
 - Target of social engineering
 - Must be well trained and informed

Components of an Information System



- Procedures
 - Written instructions for business tasks
 - Provide details on how system works
- Networks
 - Securing the data in transit
 - Block outsiders from access (firewalls)

The attackers



- The “adversary”, aka the “attacker” is a person/organization/state that actively tries the system to misbehave.
- Goals? (what they want to achieve)
 - unauthorized access, service disruption, data theft, data manipulation, reputation damage, sabotage
- Motives? (but why they act)
 - money, espionage, hacktivism, curiosity, revenge

Characteristics of Information



Any piece of information will have one or more of the following characteristics. **Goal of information security** is to maintain and preserve these characteristics.

1) Confidentiality

- Disclosure or exposure to unauthorized individuals or system is prevented

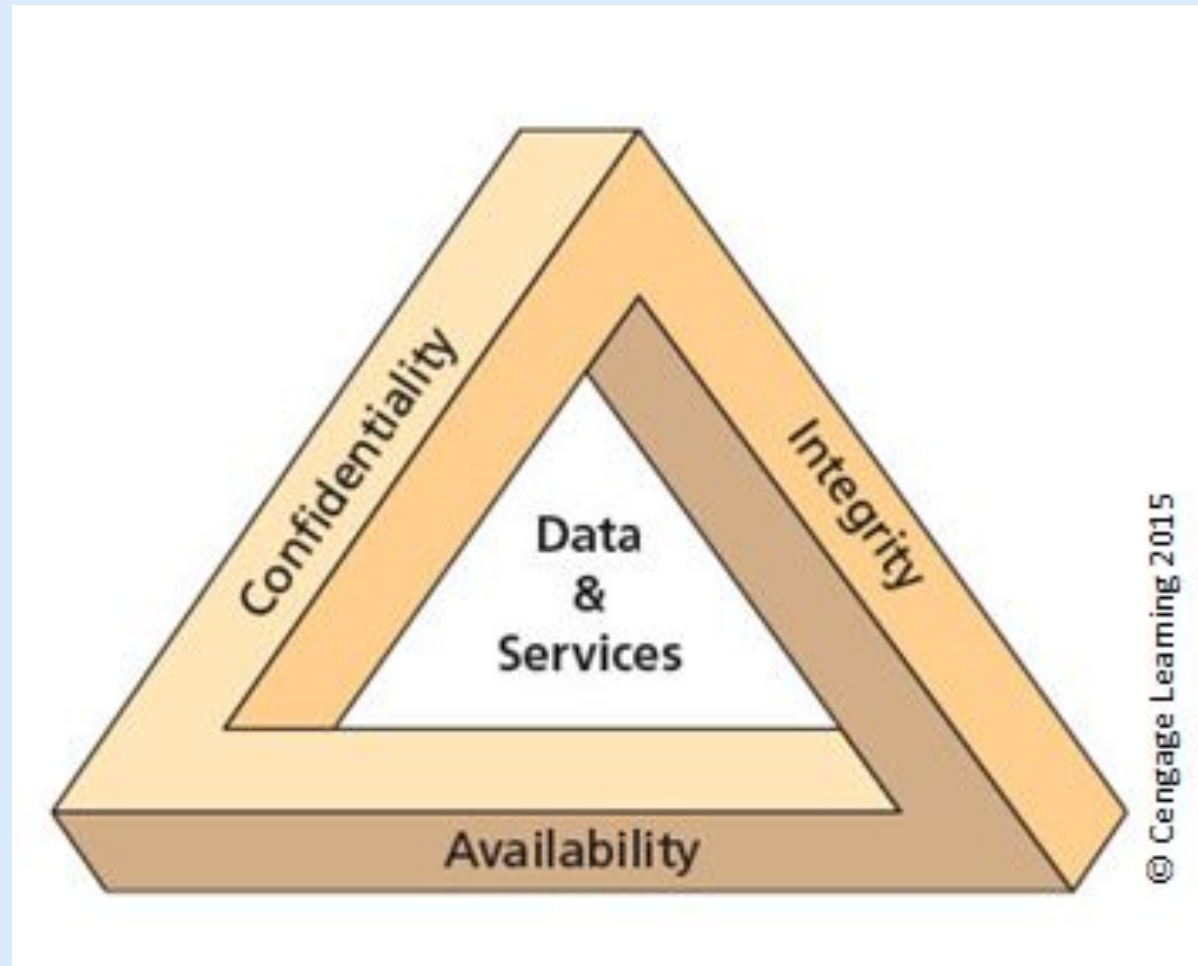
2) Integrity

- Information as a whole is complete, uncorrupted

3) Availability

- There is no interference or obstruction when information is needed

The C.I.A. triad



C, I, A are the three pillars of information security



Characteristics of Information



Some more sub-characteristics worth mentioning. All these fall under one of CIA.

Accuracy

- Information is factual, free from errors

Authenticity

- Data, communications, or entities are genuine or come from trusted source, e.g., sender of an email, creator of a document

Utility

- Data or resource remain useful, e.g. by being in a readable format/encoding and being timely
 - no value if it is too late

Characteristics of Information



Accountability

- Every action in the system can be traced back to a specific user or entity.
- Helps in after-attack recovery and finding the culprit

Privacy

- Individual's Personally Identifiable Information (PII) is collected, used and disclosed only with proper authorization and consent.

Threat vs Attack



- **Vulnerability** : weakness in an asset
- **Threat** : A potential security harm to an asset, caused by exploiting the vulnerability
- **Attack** : An act exploiting a vulnerability. Threat being realized.

Threats to information security



A house has a broken lock on the front door that the owner never got around to fixing. In the same neighborhood, a burglar is on the lookout for easy targets. One night, the burglar notices the weak lock, manages to get inside, and steals valuable items from the house.

- System has a **vulnerability** (?)
- One or more **threats** exist (?)
- An **attack** happens eventually (?)

Threats to information security



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- System has a **vulnerability** (broken lock).
- One or more **threats** exist (burglar).
- An **attack** happens eventually (stealing).

Exercise



What specific information security property is violated in the following scenarios?

	Loss of
A medical device displays incorrect dosage instructions because its software update was corrupted by poor Internet connection.	
A weather forecasting system reports rain tomorrow, but the data fed into it had mistakes, leading to wrong prediction.	
A fitness app sells users' location and health data to advertisers without their consent.	
An online shopping site shows the wrong price for an item because an attacker tampered with its database.	

Exercise



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Exercise



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	Loss of
An employee receives an email that looks like it is from their CEO, but it was actually sent by an attacker pretending to be them.	
A company cannot find out who made unauthorized changes to critical financial records because logging was disabled.	
During an online exam, a student uses someone else's credentials to log in and take the test.	
A company's internal emails are intercepted and read by a competitor while being transmitted.	

Exercise



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A company cannot find out who made unauthorized changes to critical financial records because logging was disabled.	accountability
During an online exam, a student uses someone else's credentials to log in and take the test.	authenticity
A company's internal emails are intercepted and read by a competitor while being transmitted.	confidentiality